

- 1 The total shipping cost of a purchase order will be automatically calculated by the system when a customer finalises and checks out his/her purchase from an online supermarket. The total shipping cost is the sum of the **two** components given below:
- standard shipping,
 - distance-based surcharge.

The total shipping cost for a purchase order can be calculated based on the following process:

Calculation of standard shipping

If the total amount transacted in a purchase order is more than \$150, and the user selects the mode of payment to be "PayLor!", then the standard shipping will be waived. Given that the purchase order of a customer is at most \$150, or the customer does not choose "Paylor!" as the mode of payment, a standard shipping cost of \$10 will be charged.

Calculation of distance-based surcharge

If the delivery address of the purchase order is not more than a radius of 10km from the dispatch center, no distance-based surcharge will be applied. If the destination of the purchase order stretches beyond 10km radius from the dispatch center, a distance-based surcharge of \$8 will be applied.

- (a) Create a decision table to calculate the total shipping cost for a purchase order, showing all the possible conditions and actions. [4]
- (b) Simplify your decision table by removing redundancies. [2]

- 2 (a) The original ASCII code uses 7-bits to represent **one** character.
- (i) State the total number of unique symbols that the 7-bit ASCII code can represent. [1]
- (ii) State and explain **one** shortcoming of the 7-bit ASCII code. [2]

The table below shows a list of denary, hexadecimal, and 7-bit ASCII characters.

DEN	HEX	CHAR		DEN	HEX	CHAR		DEN	HEX	CHAR		DEN	HEX	CHAR
65	41	A		78	4E	N		97	61	a		110	6E	n
66	42	B		79	4F	O		98	62	b		111	6F	o
67	43	C		80	50	P		99	63	c		112	70	p
68	44	D		81	51	Q		100	64	d		113	71	q
69	45	E		82	52	R		101	65	e		114	72	r
70	46	F		83	53	S		102	66	f		115	73	s
71	47	G		84	54	T		103	67	g		116	74	t
72	48	H		85	55	U		104	68	h		117	75	u
73	49	I		86	56	V		105	69	i		118	76	v
74	4A	J		87	57	W		106	6A	j		119	77	w
75	4B	K		88	58	X		107	6B	k		120	78	x
76	4C	L		89	59	Y		108	6C	l		121	79	y
77	4D	M		90	5A	Z		109	6D	m		122	7A	z

- (iii) The original 7-bit ASCII is extended, and it is now encoded in 8-bit instead. With workings shown clearly, find the binary codes of the characters 'A' and 'Z'. [2]
- (iv) State **one** advantage that unicode has over ASCII. [1]

(b) A MAC address expressed in 48-bit binary is:

00111000011010001001001110111011100111000000001₂

Express the MAC address given above in hexadecimal. [3]

3 A binary tree T stores the algebraic expression

$$A - B * C / D$$

It is given that T is implemented using **three** 1-D arrays `data`, `leftPtr`, and `rightPtr` as illustrated in the diagram below.

Index	data	leftPtr	rightPtr
1	'B'	0	0
2	'C'	0	0
3	'-'	4	5
4	'A'	0	0
5	'/'	6	7
6	'*'	1	2
7	'D'	0	0

root = 3

(a) Define what a binary tree is. [2]

(b) Draw the tree diagram of T . [3]

The pseudocode of a recursively-defined procedure P that takes in an integer, `index` as its parameter is given below.

```

PROCEDURE P(index : INTEGER)
  IF leftPtr[index] > 0
    THEN P(leftPtr(index))
  ENDIF
  IF rightPtr[index] > 0
    THEN P(rightPtr(index))
  ENDIF
  PRINT(data[index])
ENDPROCEDURE

```

(c) What is a recursively-defined procedure? [2]

(d) Explain the difference between a recursive and an iterative subroutine. [2]

(e) State the data structure that is used to execute a recursive function, and explain why this data structure is required in order for P to run properly. [3]

(f) Given that $P(3)$ is called in the main program, state

(i) the number of times recursive procedure P calls itself, and [3]

(ii) the final output printed when P fully completes its execution. [1]

(g) What is the purpose of procedure P ? [1]

- 4** A stack S is represented in the memory globally by an array ar , and **two** integer variables $maxSize$ and $topPointer$ that keep track of the maximum number of items S can store, and the location of the most recent item added into the ar respectively.
- (a) Describe the stack data structure. [3]
- (b) State and explain **one** advantage and **one** disadvantage of using array in the implementation of stack. [2]
- (c) State **one** application of a stack. [1]
- Given that ar has a size of 10 and $topPointer = -1$ when the stack is empty,
- (d) Write down the pseudocode for procedure $Push(item)$ and function $Pop()$. [4]
- 5** A hash table h is a type of data structure that stores key-value pairs. It is given that h uses chaining to handle collisions, and h is implemented using an array of list.
- (a) What is a key-value pair? [1]
- (b) Describe how key-value pairs are stored in a hash table. [3]
- (c) What is a collision? [1]
- (d) State and explain one benefit of implementing h with an array of list. [2]

- 6** The Singapore Dragon Boat Festival (SDBF) is an annual two-day dragon boat racing festival that is made up of many race events. The event organiser will use the following information to record data related to the race events, race results, participating dragon boat teams and paddlers.

Every dragon boat team is represented by a team manager. The role of the team manager is to register his/her club's paddlers to the SDBF race events, and also to keep track of the team's results of the race events. A team manager is the point-of-contact between the event organiser and the team, and he/she does not participate in any race event. Each dragon boat team has a unique name, and the name, contact number and email address of the team manager are recorded by the event organiser.

A paddler can only represent one dragon boat team, but he/she can represent the team in more than one race event throughout the 2-day SDBF. In addition, a dragon boat team is not allowed to send more than one team of paddlers to participate in a race event. The event organiser will issue every paddler with a unique `paddlerID` when the `name`, `gender`, `contactNo`, and `address` of a paddler are registered.

The information of every race event in this two-day festival will be recorded and managed. It is given that each race event has a unique `raceID`, `raceCategory`, `raceDistance`, `raceDate` and `raceTime` will be stored by the event organiser.

- (a)** Draw the E-R diagram showing the tables and the relationships between them. [4]

- (b)** The race organiser has decided to computerise and use a relational database to store and manage all the information related to SDBF. Given that a table description can be expressed as:

`TableName (Attribute1, Attribute2, ...)`

The primary key is indicated by underlining **one** or more attributes, while the foreign keys are indicated by using asterisks.

Write down the table descriptions for the tables when they are in fully normalised form, and state the key references of the foreign keys identified. [5]

- (c) In order to remind paddlers to report for their race events punctually, the event organiser has decided to issue a personal race schedule to paddlers who request for one. Write down the Structured Query Language (SQL) that takes in the input `paddlerID` and returns the race schedule of that paddler shown in the table below. [4]

ID of Paddler: 103

RaceID	RaceCategory	RaceDate	RaceTime	RaceDist
1	22-Men Crew	06/06/22	0800	200m
4	10-Men Crew	06/06/22	0900	200m
21	10-Mixed Crew	07/06/22	0930	500m

The event organiser engages a software company to develop a program to manage the information of the festival. The software company has decided to use object-oriented programming to implement this program, and is suggesting a change to have the `Paddler` and `Team Manager` classes inherit from the `Personnel` class. Also, `Paddler` class and `RaceEvent` class share a HAS-A relationship with each another.

- (d) Draw a class diagram that shows the following for the context described above. [8]
- The superclass.
 - Any subclasses.
 - Inheritance and encapsulation.
 - Attributes.
 - Appropriate methods.
- (e) State the purpose of a superclass. [2]

- 7 The pseudocode below describes an algorithm that searches for a person's name that is stored in an array, `ar` of `N` unique names.

```

1  DECLARE first, mid, last : INTEGER
2  DECLARE name: STRING
3  name ← INPUT("Enter search name:")//prompts for a STR input
4  first ← 1
5  last ← N
6  REPEAT
7      mid ← (first+last) DIV 2 //DIV 2 returns integer quotient
8      IF _____ THEN
9          first ← mid + 1
10     ELSE
11         last ← mid - 1
12     ENDIF
13 UNTIL first > last or ar[mid] = name

```

- (a) What is the name of the search algorithm above? [1]
- (b) What condition has to be satisfied in order for the search algorithm to work? [1]
- (c) Explain how a REPEAT-UNTIL loop is different from a WHILE loop. [2]
- (d) Fill in the blank in line 8 of the pseudocode. [1]
- (e) Suggest one method to improve code readability. [1]
- (f) Given that $N = 168$, and $ar[10] = \text{"CLEMENT"}$. With workings or diagrams shown, find the number of iterations performed by the search algorithm before the name "CLEMENT" is found. [3]

An array contains the following integers:

20 49 13 52 27 97 12 42 23 30

- (g) Describe, with the aid of a diagram, how merge sort can be used to sort the integers in the array above in ascending order. [5]
- (h) The merge sort is said to have the same time complexity for all 3 cases (best, average, and worst). [1]
- (i) State the time complexity of merge sort. [1]
- (ii) Explain why the time complexities for all cases of merge sort are the same. [2]

- 8 (a) What is a wide area network (WAN)? [1]
- (b) Define communication protocol, and explain why it is necessary in the context of data transmission over WAN. [3]
- (c) Describe how data packets travel across WAN sent from one device and received by another device. [3]
- 9 Mdm Liu spent more than \$100 in Happy Supermarket and was entitled to take part in a lucky draw contest with an opportunity to win a Volvo CX40 car. In order to take part in the lucky draw, Mdm Liu had to fill up her personal particulars such as the last 4 digits and character of her identity card number, her residential address, and her personal contact number. The lucky draw took place and Mdm Liu did not win any prize. Days later, Mdm Liu received numerous calls from sales representatives from companies unbeknownst to her. Mdm Liu reflected that she had not shared her contact details with any businesses or sales representatives in recent days.
- (a) What could be the reason for the unexpected calls made to Mdm Liu? [1]
- (b) State and explain **two** aspects from the Singapore Personal Data Protection Act (PDPA) that aims to govern the collection, use, disclosure and care of personal data. [4]